MathQuiz #2

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1 CORRECTION NOTES

This is now a corrected copy of my quiz. Because of that, I cut out parts that weren't necessary for deriving the answer for the problem, even if they showed engagement with the problems.

2 Question One

a)
The term s_n given the polynomials listed can be written as:

$$(S_n) = \frac{2n+3}{3n^2+4n+5}$$

Before we attempt to find $lim_n(s_n)$, it would be wise to make sure that there IS a limit. If we plug in S_n from $n = 0$ to $n = 10$, the followed values are our outputs are:

$$0.6, 0.417, .28, .205, 0.13, 0.109, 0.094, 0.083, 0.074, 0.067$$

From intuition, it seems reasonable assume there is a limit as the values are slowly approaching something.
Now being a little more confident that a limit exist, we can find $lim_n(s_n)$ For convenience, I am going to divide the fraction by the highest power over itself so that we can use the limits from *Prop* 3.2.8. In this case, we are dividing by $\frac{n^2}{n^2}$

$$\frac{2n+3}{3n^2+4n+5} \div \frac{n^2}{n^2} = \frac{\frac{2}{n}+\frac{3}{n^2}}{3+\frac{4}{n}+\frac{5}{n^2}}$$

I divided by n^2/n^2 first due to convenience, but the reason we can find the limit of the fraction is that the quotient rule means

$$lim_n(s_n) = lim_n(\frac{2}{n}+\frac{3}{n^2}) \div \ \ lim_n(3+\frac{4}{n}+\frac{5}{n^2})$$

Now that we can divide the top limit by the bottom limit, we now just have to find the limits

$$lim_n(\frac{2}{n} + \frac{3}{n^2}) = (0 + 0) = 0$$

This is because $\frac{2}{n}$ goes to 0 as per *Prop* 3.2.8 #2 and $\frac{3}{n^2}$ goes to 0 because of *Prop* 3.2.8 #3

$$lim_n(3 + \frac{4}{n} + \frac{5}{n^2}) = 3 + 0 + 0 = 3$$

This is because 3 goes to 3 as per the *Prop* 3.2.8 #1, or the constant rule. The other two monomials have limits of 0 for same exact reasons as above for $\frac{2}{n}$ and $\frac{3}{n^2}$ These calculations leave us with

$$lim_n(s_n) = lim_n\left(\frac{\frac{2}{n} + \frac{3}{n^2}}{3 + \frac{4}{n} + \frac{5}{n^2}}\right) = \frac{0}{3} = 0$$

$$lim_n(s_n) = 0$$

This limit is supported by previous data which seems to show the sequence s_n approaching 0
t_n can defined from the polynomials as *(substitution slip fixed)*

$$t_n = \frac{\frac{1}{2n} + 3}{\frac{3}{n^2} + \frac{1}{4n} + 5}$$

In a phenomenal show of convenience, the terms are already in easy form to find limits of, so no need to divide by a term. Assuming the quotient rule, can find the limits of each term using *Prop* 3.2.8 #1, #2, and #3 (although I explained in detail which prop applied to which term for previous limit, since all of the propositions are the same for this limit and are applied the exact same, I am going to leave the explanation out)

$$lim_(t_n) = lim_n\left(\frac{\frac{1}{2n} + 3}{\frac{3}{n^2} + \frac{1}{4n} + 5}\right) = \left(\frac{0 + 3}{0 + 0 + 5}\right) = \left(\frac{3}{5}\right)$$

However, we should still check that this sequence is converging. The terms for t_n from $n = 1$ ($n = 0$ is undefined since some terms in the function are $\frac{1}{n}$) to $n = 5$ are:

$$0.627, 0.624, 0.618, 0.615, .612$$

The terms seems to converge to a value, and 0.6 seems like a perfect value. So, $lim_n(t_n)$ exist and

$$lim_n(t_n) = \frac{3}{5}$$

u_n when defined with given polynomials is

$$u_n = \frac{2n^2 + 3}{3n^2 + 4n + 5}$$

Since the terms aren't "pretty" in context of finding the limit, will do same as done for first limit and divide by largest term, $\frac{n^2}{n^2}$

$$\frac{2n^2+3}{3n^2+4n+5} \div \frac{n^2}{n^2} = \frac{2+\frac{3}{n^2}}{3+\frac{4}{n}+\frac{5}{n^2}}$$

Now finding limit assuming quotient rule, and using *Prop* 3.2.8 #1, #2, and #3, can find limit of u_n

$$lim_n(u_n) = lim_n\left(\frac{2+\frac{3}{n^2}}{3+\frac{4}{n}+\frac{5}{n^2}}\right) = \left(\frac{2+0}{3+0+0}\right) = \left(\frac{2}{3}\right)$$

Doing a reality check from $n = 0$ to $n = 5$ and also checking $n = 200$ and $n = 1000$ gives us

$$0.417, 0.44, 0.477, 0.507, 0.53, 0.662, 0.666$$

As said above last two terms are $n = 200$ and $n = 1000$. This sequence seems to converge and the limit of $\frac{2}{3}$ seems obvious. So, $lim_n(u_n)$ exist and

$$lim_n(u_n) = \frac{2}{3}$$

b) This problem is similar to the homework problem 3.2.14 e, except that both power functions are variables and can be different.

$$lim_n\left(\frac{(3+\frac{1}{n})^\alpha - 3^\alpha}{(4+\frac{1}{n})^\beta - 4^\beta}\right)$$

The above limit's terms can be represented by power functions (similar to e from homework)

$$lim_n\left(\frac{(3+\frac{1}{n})^\alpha - 3^\alpha}{(4+\frac{1}{n})^\beta - 4^\beta}\right) = lim_n\left(\frac{f_\alpha(3+\frac{1}{n}) - f_\alpha(3)}{f_\beta(4+\frac{1}{n}) - f_\beta(4)}\right)$$

The next that follows from when we did problem e was to match

$f_\alpha(3+\frac{1}{n}) - f_\alpha(3)$ with $3+\frac{1}{n}-3$

$f_\beta(4+\frac{1}{n}) - f_\beta(4)$ with $4+\frac{1}{n}-4$

The reason for this is that with $\frac{f(a)-f(x)}{a-x}$, as $a \to x$, $\frac{f(a)-f(x)}{a-x} \to f\prime(x)$. This means we can turn the power functions into an easy to find derivative by having the inverse of each of the above power functions be its input. Something else that makes our lives easier is that $(3+\frac{1}{n})-3) = \frac{1}{n}$ and $(4+\frac{1}{n}-4) = \frac{1}{n}$. Because each constant is added and subtracted, we can multiply the equation by

$$\left(\frac{(4+\frac{1}{n})-4}{(3+\frac{1}{n})-3)}\right)$$

Because of the above observation, the above helper is still just 1. So if we multiply the original term by our helper

$$lim_n \left(\frac{f_\alpha(3+\frac{1}{n}) - f_\alpha(3)}{(3+\frac{1}{n}) - 3)} * \frac{(4+\frac{1}{n}) - 4}{f_\beta(4+\frac{1}{n}) - f_\beta(4)} \right)$$

Since $\frac{f(a)-f(x)}{a-x}$ is satisfied for both (the right term is the inverse)

$$lim_n \left(\frac{f_\alpha\prime(3)}{f_\beta\prime(4)} \right)$$

Derivative function is $f_n\prime(x) = nf(x)^n - 1$, so can change the two primes functions to

$$lim_n \left(\frac{\alpha 3^{\alpha-1}}{\beta 4^{\beta-1}} \right)$$

Reason for 3 and 4 is $\frac{f(a)-f(x)}{a-x}$ establish $f(x) = 3$ for term with α and that $f(x) = 4$ for the term with β.
Because both $\alpha 3^{\alpha-1}$ and $\beta 4^{\beta-1}$ are constants, and constants are their own limit per *Prop* 3.2.8 #1, this cannot be simplified further. Since both numerator and denominator are limits, we can find lim_n of the problem by using quotient rule. That is to say

$$lim_n \left(\frac{(3+\frac{1}{n})^\alpha - 3^\alpha}{(4+\frac{1}{n})^\beta - 4^\beta} \right) = lim_n \left(\frac{\alpha 3^{\alpha-1}}{\beta 4^{\beta-1}} \right) \text{ and exist}$$

3 QUESTION Two

a) Since this is just calculations, won't do any text explanation

$$s_1 = 2$$

$$s_{1+1} = \frac{s_1^2 + 1}{2s_1 - 1} \rightarrow s_2 = \frac{2^2 + 1}{2(2) - 1} = \frac{5}{3}$$

$$s_{2+1} = \frac{s_2^2 + 1}{2s_2 - 1} \rightarrow s_3 = \frac{\frac{5}{3}^2 + 1}{2\frac{5}{3} - 1} = \frac{34}{21}$$

$$s_1 = 2, s_2 = 1.666, s_3 = 1.619$$

b) Since we are finding the limit of the equation, can write

$$lim_n(s_n + 1) = \frac{{s_n}^2 + 1}{2s_n - 1}$$

However, index translation doesn't matter (*Prop* 3.2.13), can say

$$lim_n(s_n) = \frac{{s_n}^2 + 1}{2s_n - 1}$$

Since it's convenient for the problem, let

$$\ell = lim_n(s_n)$$

Can substitute ℓ

$$\ell = \frac{\ell^2 + 1}{2\ell - 1}$$

Using quotient rule, sum/difference rule, can solve

$$2\ell^2 - \ell = \ell^2 + 1 = \ell^2 - \ell - 1$$

Can solve for ℓ with quadratic formula

$$\ell = \frac{1 \pm \sqrt{1^2 - 4(1)(-1)}}{2 * 1} \rightarrow \frac{1 \pm \sqrt{5}}{2}$$

Using the approximation of $\sqrt{5} = 2.236$, two possible ℓ are

$$\frac{1 + 2.236}{2} = 1.618$$

$$\frac{1 = 2.236}{2} = -0.618$$

As stated in problem, s_n is positive, thus the limit will be the positive option. Our terms from a were $s_1 = 2, s_2 = 1.666, s_3 = 1.619$. Because of that, $\ell = 1.618$ makes sense since they seem to rapidly converge to the limit.

4 QUESTION Three

GIVEN:

$$1 : S_n \leq A \leq T_n \text{ for all positive integer } n \tag{3.1}$$

$$2 : lim_n(T_n - S_n) = 0 \text{ and exist} \tag{3.2}$$

for all real $n \geq 1$ Given 3.1, we can subtract S_n from all

$$S_n - S_n \leq A - S_n \leq T_n - S_n = 0 \leq A - S_n \leq T_n - S_n$$

$$0 \leq A - S_n \leq T_n - S_n \text{ for each n} \tag{3.3}$$

We want to use Squeeze rule, and since 3.2 already proves the right limit, all we need to do is prove

$$lim_n(0) = 0 \tag{3.4}$$

3.4 is proven by *Prop* 3.2.8 #1, so we can use squeeze rule to say

$$lim_n(A - S_n) = 0 \text{ and exist} \tag{3.5}$$

Clever use of identity property will allow us to 3.5 to great effect

$$S_n = S_n \text{ for each } n$$

$$S_n = S_n + A - A \text{ by Identity property for each } n$$

$$S_n = A + [S_n - A] \text{ for each } n$$

Per 3.5 and Constant rule(*Prop* 3.2.8 #1 proves $lim_n(A) = A$), and with use of Sum and Difference rules,

$$lim_n(S_n) = A + 0$$

$$lim_n(S_n) = A \text{ and exist} \tag{3.6}$$

Using the same principle, we will cleverly prove the same for T_n using 3.6

$$T_n = T_n \text{ for each } n$$

$$T_n = T_n - S_n + S_n \text{ by Identity property for each } n$$

$$T_n = [T_n - S_n] + S_n \text{ for each } n$$

Per 3.2 and 3.6 with use of Sum and Difference rules,

$$lim_n(T_n) = 0 + A$$

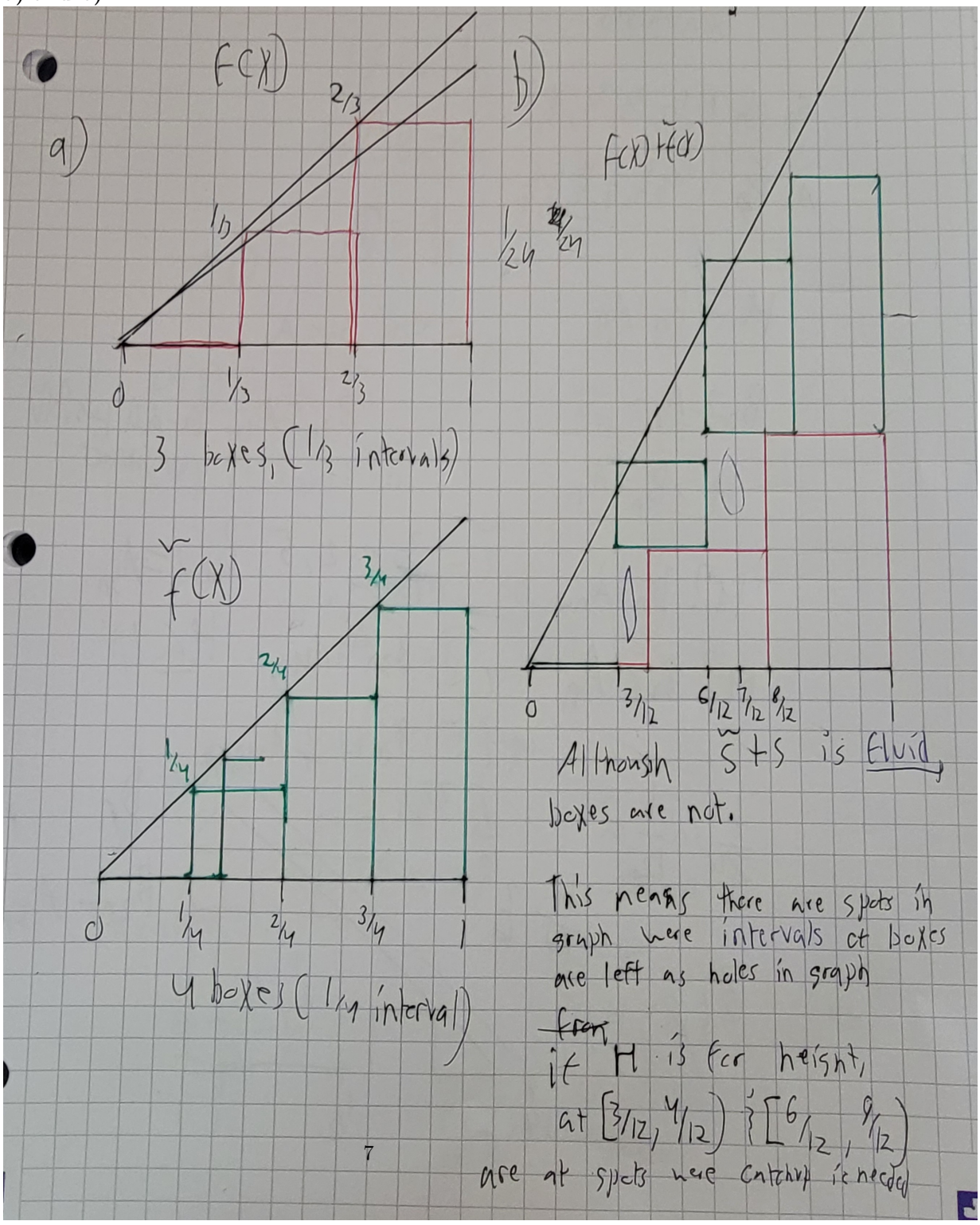
$$lim_n(T_n) = A \text{ and exist} \tag{3.7}$$

Combining 3.6 and 3.7 we now have our desired definition, given 3.1 and 3.2 are true that

$$lim_n(T_n) = lim_n(S_n) = A \text{ and both } lim_n(T_n) \text{ and } lim_n(S_n) \text{ exist}$$

5 QUESTION Four

a) and b)



c) I couldn't find a solution that I felt would work, I was mainly trying add together the function of the two sets of boxes in a way where there weren't holes created.
Such that boxes for $f(x)$ are represented by A_n) and $\tilde{f}(x)$ is by B_n. Let n be all natural numbers, such that n represents the intervals of the boxes.
Sequence $A_n = \frac{n}{a}$ where a is the number of boxes in $f(x)$, let same apply so that $B_n = \frac{n}{b}$ such that b is the number of boxes in $\tilde{f}(x)$. Also let all $B_n \geq A_n$.
For example, B_n would be $B_2 = \frac{2}{4}$, and A_n would be $A_1 = \frac{1}{3}$. Issue with current function is that at certain points for $f(x) + \tilde{f}(x)$ have extra interval(n) of B_n. For example, $f + \tilde{f}(6/12) = 1$, but interval of their boxes are messed up as illustrated in the previous picture, resulting in $B_2 + A_2$ instead of what should be $B_1 + A_2$. This results in the height of $A_n + B_n$ at $\frac{6}{12}$ to be $\frac{14}{12}$ instead of what it should be (and what $S_n + \tilde{S}_n$ is), $\frac{10}{12}$.
From this point I was lost and didn't know how to continue.